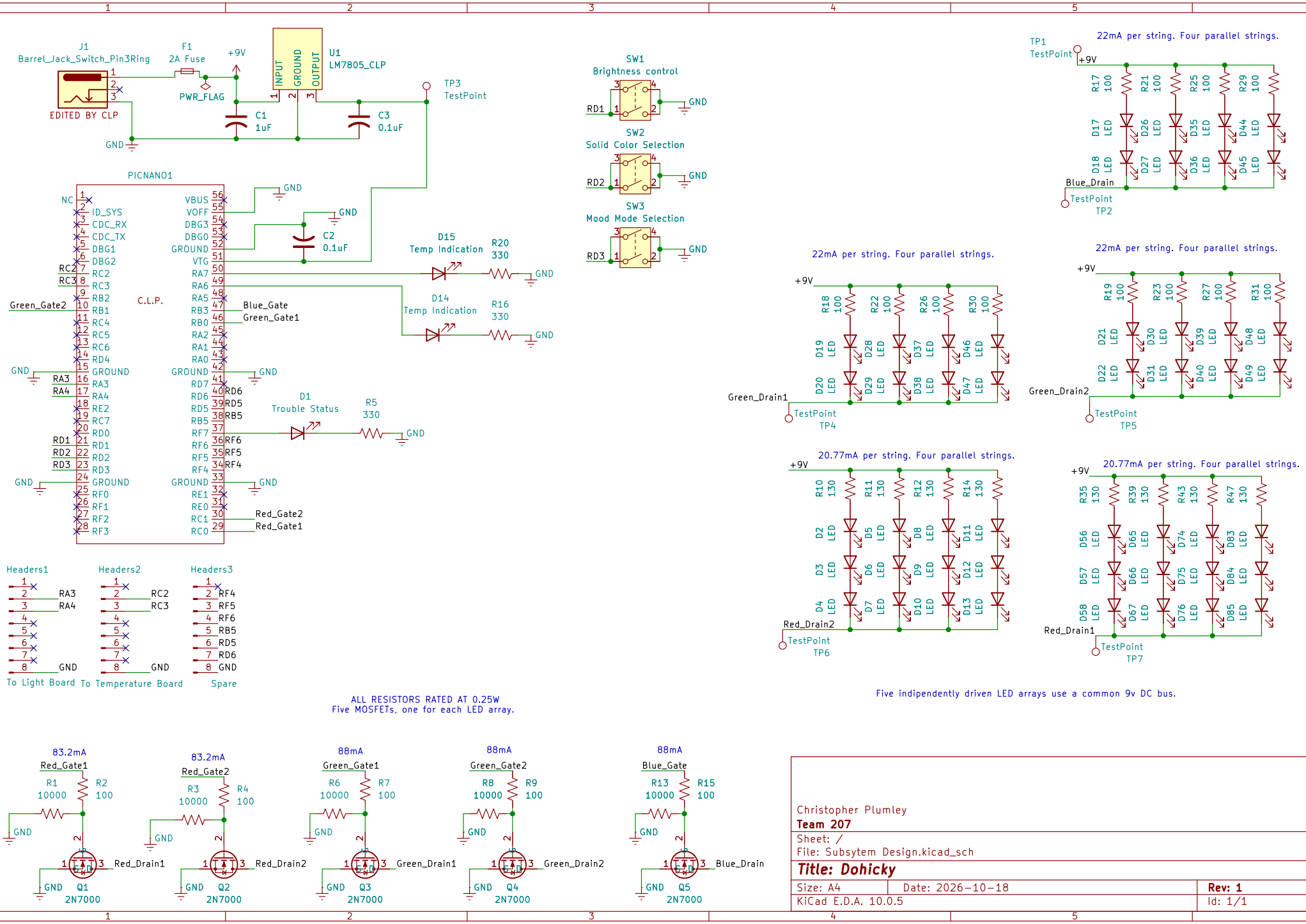

J1
Barrel_Jack_Switch_Pin3Ring
EDITED BY CLP
F1
2A Fuse
PWR_FLAG
+9V
U1
LM7805_CLP
INPUT
GROUND
OUTPUT
C1
1uF
C3
0.1uF
TP3
TestPoint
GND
PICNANO1
C.L.P.
C2
0.1uF
D15
Temp Indication
R20
330
D14
Temp Indication
R16
330
D1
Trouble Status
R5
330
Green_Gate2
Green_Gate1
Blue_Gate
Red_Gate2
Red_Gate1
SW1
Brightness control
SW2
Solid Color Selection
SW3
Mood Mode Selection
RD1
RD2
RD3
Headers1
Headers2
Headers3
To Light Board
To Temperature Board
Spare
ALL RESISTORS RATED AT 0.25W
Five MOSFETs, one for each LED array.
83.2mA
Red_Gate1
R1
10000
R2
100
Red_Drain1
Q1
2N7000
83.2mA
Red_Gate2
R3
10000
R4
100
Red_Drain2
Q2
2N7000
88mA
Green_Gate1
R6
10000
R7
100
Green_Drain1
Q3
2N7000
88mA
Green_Gate2
R8
10000
R9
100
Green_Drain2
Q4
2N7000
88mA
Blue_Gate
R13
10000
R15
100
Blue_Drain
Q5
2N7000
22mA per string. Four parallel strings.
TP1
TestPoint
Blue_Drain
TP2
22mA per string. Four parallel strings.
Green_Drain1
TP4
22mA per string. Four parallel strings.
Green_Drain2
TP5
20.77mA per string. Four parallel strings.
Red_Drain2
TP6
20.77mA per string. Four parallel strings.
Red_Drain1
TP7
Five indipendently driven LED arrays use a common 9v DC bus.
Christopher Plumley
Team 207
Sheet: /
File: Subsytem Design.kicad_sch
Title: Dohicky
Size: A4
Date: 2026-10-18
Rev: 1
KiCad E.D.A. 10.0.5
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